

State of Health Prediction Method for Ship Propulsion Batteries Based on a Parallel TCN-LSTM Architecture

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Abstract—This study proposes a State of Health (SOH) prediction method for ship propulsion batteries. First, three health-related features are extracted from battery charging voltage curves. Next, a Temporal Convolutional Network (TCN) and a Long Short-Term Memory (LSTM) network are constructed to capture the complex nonlinear relationships between these features and battery SOH, producing high-dimensional feature representations. Finally, a cross-attention mechanism fuses the output features of the TCN and LSTM into a parallel TCN-LSTM model, thereby improving SOH prediction accuracy. Experimental results demonstrate that the proposed TCN-LSTM model achieves a prediction error of 0.0111—representing a 37.99% reduction compared to the standalone TCN model and a 45.32% reduction compared to the standalone LSTM model, which highlighting the superiority of the proposed approach.

Keywords—component; Battery; Ship; State of Health; TCN-LSTM

I. INTRODUCTION

The rapid advancement and application of battery technology in electric vessels have become a key driver of green shipping and sustainable development[1]. As electric vessels gain widespread adoption, propulsion batteries—as the core energy source—directly affect operational efficiency, safety, and economic performance through their state of health (SOH). Consequently, accurate prediction of battery SOH is critically important. Reliable SOH estimation enables early detection of battery faults and anomalies, optimizes maintenance and replacement schedules, extends battery lifespan, and enhances navigational safety [2]. Moreover, with continual expansion of shipboard battery system scale, traditional manual inspection and maintenance approaches no longer satisfy requirements for efficiency and precision. Therefore, development of intelligent SOH prediction methods is essential to ensure the reliability and stability of electric vessels.

Existing SOH prediction techniques for ship propulsion batteries can be broadly categorized into statistical methods,

physics-based models, and data-driven approaches. Statistical techniques—such as regression and time-series analysis—model battery SOH by fitting historical data. While these methods can reflect general health trends, they lack the capacity to capture complex nonlinear relationships inherent in battery degradation [3]. Physics-based models rely on electrochemical principles to infer SOH by simulating internal battery dynamics. Although theoretically rigorous, these models require extensive parameter calibration, involve high computational complexity, and are unsuitable for real-time prediction [4]. In contrast, deep learning methods have demonstrated superior performance in SOH estimation by automatically extracting features from high-dimensional, nonlinear data. Particularly for time-series prediction, deep neural networks effectively learn degradation patterns under varying operating conditions, yielding more accurate SOH estimates [5]. Convolutional neural networks, gated recurrent units, and long short-term memory networks have been widely applied to ship battery SOH prediction [6,7]. However, single-model approaches often exhibit limited generalizability across diverse datasets and operating scenarios.

To address these limitations, this study proposes a parallel CTN-LSTM architecture for ship propulsion battery SOH prediction. First, a TCN is employed for efficient extraction of local temporal features. Next, an LSTM network captures long-term dependencies in SOH degradation. Finally, a cross-attention mechanism fuses the complementary feature representations from both models, enhancing overall prediction accuracy.

II. DATASET AND FEATURE EXTRACTION

A. Definition of State of Health (SOH)

State of health quantifies the relationship between a battery's current performance and its design specifications. It is defined as the ratio of the battery's measured capacity to its rated capacity, as shown in Equation:

$$SOH = \frac{C_{actual}}{C_{nominal}} \times 100\% \quad (1)$$

where C_{actual} denotes the battery's current measured capacity and $C_{nominal}$ denotes its rated capacity.

B. Dataset

This study utilizes battery cycling data from the CALCE dataset, specifically three cells labeled CS35 and CS36 [8]. All cells were charged under a constant-current/constant-voltage (CC/CV) protocol with a constant-current rate of 0.5 C, a CC cutoff voltage of 4.2 V, and a CV cutoff current of 0.05 A.

C. Feature Extraction

Because ship propulsion battery discharge behavior is influenced by multiple factors and exhibits complex, irregular patterns, this study extracts features exclusively from the charging process to ensure stability and reliability. As shown in Figure 1, increasing cycle count leads to a decrease in constant-current charging time (CCCT), since batteries with lower SOH store less charge and thus reach the cutoff voltage more quickly. Conversely, constant-voltage charging time (CVCT) increases as SOH declines, reflecting reduced charging efficiency. During the constant-current phase, the overall charging voltage curve shifts upward and leftward as SOH decreases, resulting in an increase in constant-current average voltage (CCAV). CCAV is negatively correlated with SOH, indicating a decline in current-carrying capacity. Accordingly, CCCT, CVCT, and CCAV are selected as health-related features for ship propulsion battery SOH prediction.

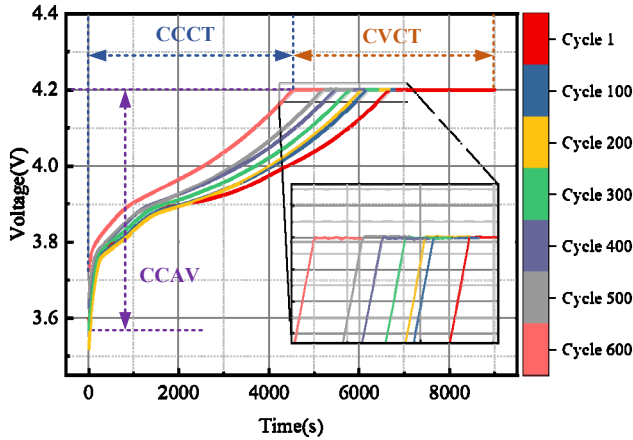


Figure 1: Feature Extraction

III. TCN-LSTM PARALLEL MODEL

A. Temporal Convolutional Network

Convolutional operations are central to the TCN architecture, implemented as causal dilated convolutions [9]. Causal convolution ensures that each output at time step depends only on current and past inputs, thereby preserving the chronological order of time-series data and preventing information from future time steps influencing the result. This is achieved through zero

padding, which prevents the convolutional kernel from accessing future data points.

Dilated convolution increases the receptive field by inserting gaps (dilations) between kernel elements. This allows the network to capture long-range temporal dependencies without requiring a deep stack of layers. The dilated convolution operation is defined as:

$$y(t) = \sum_{i=0}^{k-1} x(t + r \cdot i) \cdot w(i) \quad (2)$$

where $y(t)$ is the output at time t , $x(t)$ is the input signal, $w(i)$ denotes the convolutional kernel weights, r is the dilation factor, and k is the kernel size. Adjusting r controls the effective receptive field size.

To mitigate the vanishing gradient problem during training, the TCN architecture incorporates residual connections. A residual connection adds the input of a layer directly to its output, enabling information to bypass intermediate layers and thereby facilitating more effective gradient propagation. The residual block is defined as:

$$y = \mathcal{F}(x, \{W_i\}) + x \quad (3)$$

where x represents the input to the block, $\mathcal{F}(x, \{W_i\})$ denotes the output of the convolutional operations, and y is the block's output. By providing a shortcut path, residual connections ensure that information can be transmitted directly to subsequent layers, improving training stability and convergence. The structural components of the TCN, including the dilated convolution and residual connection modules, are illustrated in Figure 2.

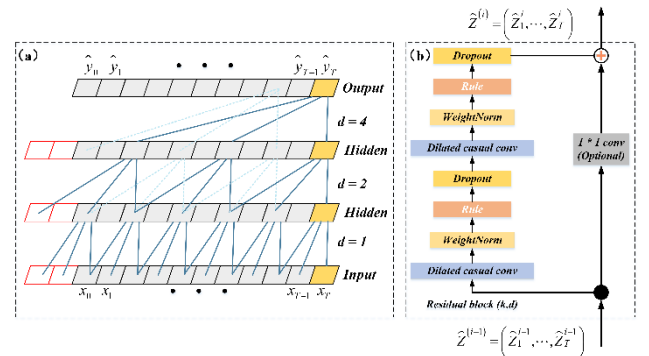


Figure 2. TCN structural elements: (a) Dilated convolution; (b) Residual connection.

B. Long Short-Term Memory Network

As illustrated in Figure 3, the Long Short-Term Memory (LSTM) network employs three gating mechanisms—forget, input, and output gates—to regulate information flow and thereby capture long-term dependencies in sequential data [10]. The forget gate determines the extent to which information from the previous hidden state and the current input should be discarded. Its operation is defined by:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

where f_t denotes the forget gate output, W_f is the forget gate weight matrix, h_{t-1} is the hidden state at the previous time step, x_t is the current input vector, and b_f is the bias term.

The input gate controls the proportion of current information that is incorporated into the cell state. Its computation comprises two components: one computes the input gate activation via a sigmoid function, and the other generates a candidate cell state via a tanh function. The output gate determines the hidden state h_t at the current time step. Its operation is defined as:

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (5)$$

where o_t denotes the output gate's output, W_o is the output gate weight matrix, and b_o is the bias term.

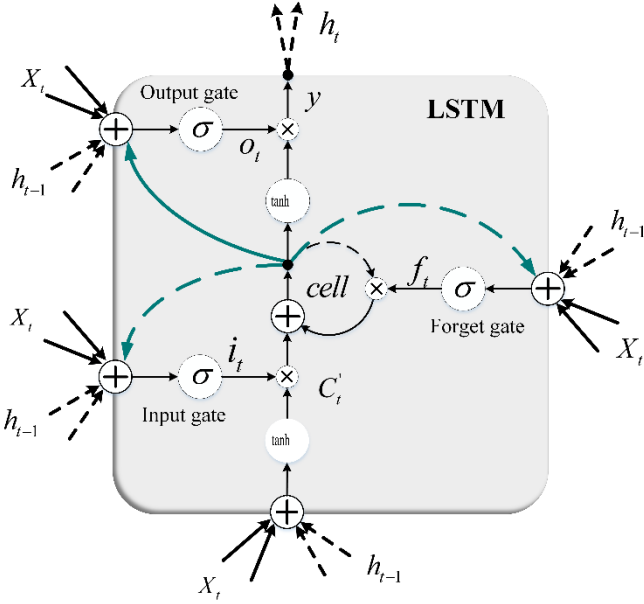


Figure 3. LSTM cell architecture.

C. TCN-LSTM Parallel Model

As illustrated in Figure 4, the proposed TCN-LSTM parallel model comprises two concurrent branches: a TCN branch and an LSTM branch. Both branches receive the three extracted health features as input. The TCN branch consists of stacked TCN layers followed by a fully connected layer, while the LSTM branch comprises one or more LSTM layers followed by a fully connected layer.

To leverage the complementary strengths of each architecture, a cross-attention mechanism fuses their high-dimensional feature representations. Specifically, the TCN output serves as the Query (Q), and the LSTM output serves as both the Key (K) and Value (V). This arrangement enables the model to weight and integrate information from the LSTM context when refining TCN-derived features. The cross-attention operation is defined as:

$$\text{Cross-Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (6)$$

In Equation, Q denotes the Query, K the Key, and V the Value; d_k represents the dimensionality of the Key vectors. The softmax function normalizes the attention scores, while the scaled dot-product computes the similarity between Query and Key. By employing this cross-attention mechanism, the TCN branch dynamically reweights its output based on the contextual information captured by the LSTM branch, thereby improving its capability to model the dynamic temporal patterns in the data.

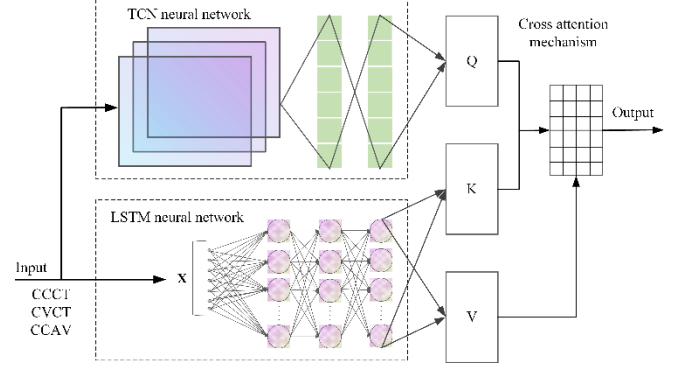


Figure 4. Parallel TCN-LSTM architecture for State of Health prediction.

IV. RESULTS AND DISCUSSION

The effectiveness of the proposed TCN-LSTM parallel model was evaluated using data from battery cells CS35 and CS36. For each cell, the first 30% of the cycling data was used as the training set, and the remaining 70% as the test set. To demonstrate the superiority of the parallel architecture, its performance was compared against standalone TCN and standalone LSTM models. Prediction accuracy was assessed using four metrics: root mean square error (RMSE), mean absolute percentage error (MAPE), mean absolute error (MAE), and the coefficient of determination (R^2).

Figures 5 and 6 depict the SOH prediction curves for cells CS35 and CS36 using the different models. The TCN-LSTM model's predictions most closely match the ground-truth SOH trajectory, exhibiting near-perfect alignment during the early cycles. In the later cycles—characterized by a rapid SOH drop and limited training data—the TCN-LSTM predictions show minor deviations from the actual values. Overall, the parallel TCN-LSTM model achieves the best fit to the SOH curves, whereas the standalone TCN and LSTM models demonstrate substantially poorer performance.

Figures 7 and 8 present the error scatter plots, showing that the scatter plot of the TCN-LSTM model is closer to the diagonal line, indicating smaller prediction errors. The prediction errors of the TCN-LSTM model are distributed on both sides of the diagonal, while those of the standalone TCN and LSTM models are mostly positive, suggesting that the TCN-LSTM model has a more balanced and reasonable error distribution. As shown in Table 1, the RMSE values for the TCN-LSTM, TCN, and LSTM models are 0.0111, 0.0179, and 0.0203, respectively. The TCN-LSTM model achieved the lowest error, reducing RMSE by

37.99% and 45.32% compared to the TCN and LSTM models, respectively. These results demonstrate the superiority of the

proposed method and provide a valuable reference for SOH prediction of marine power batteries.

Table 1 Prediction Errors for SOH Estimation

Battery	Model	Error index			
		RMSE	MAPE	MAE	R ²
1	TCN-LSTM	0.0105	1.01%	0.0084	0.9615
	TCN	0.0176	1.54%	0.0124	0.8356
	LSTM	0.0199	1.76%	0.0142	0.7906
CS36	TCN-LSTM	0.0117	1.12%	0.0089	0.9690
	TCN	0.0181	1.62%	0.0124	0.9268
	LSTM	0.0206	1.86%	0.0142	0.9044

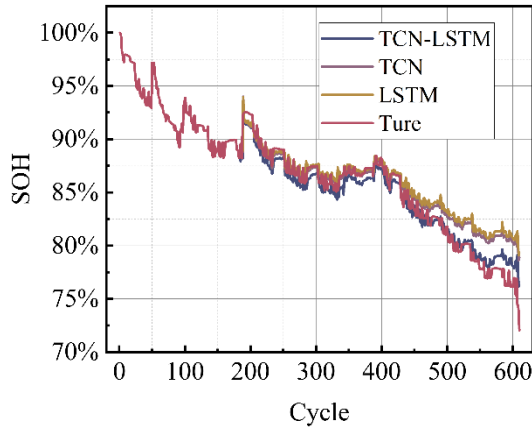


Figure 5. Prediction Curve of CS35

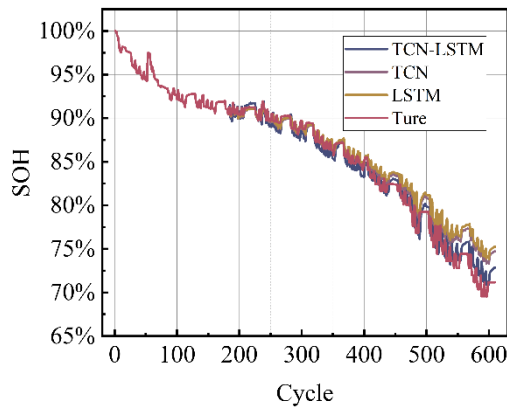


Figure 6. Prediction Curve of CS36

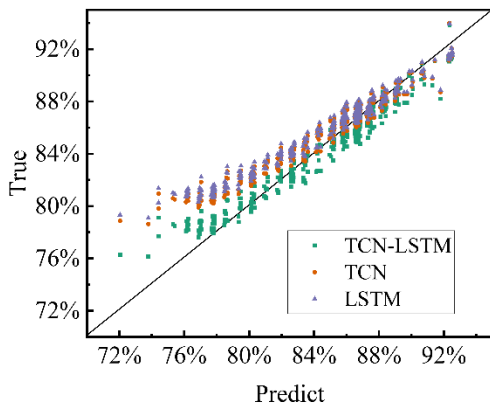


Figure 7. Prediction Error of CS35

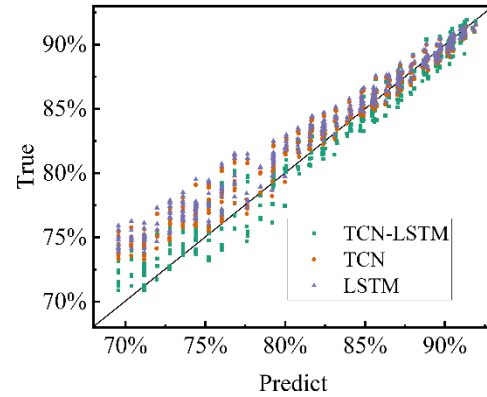


Figure 8. Prediction Error of CS36

V. CONCLUSIONS

This study proposed a TCN-LSTM parallel architecture-based method for predicting the State of Health of marine power batteries. The main work and conclusions are as follows:

(1) The charging voltage curves were analyzed, and key health features were extracted, including constant current charging time, constant voltage charging time, and average voltage during constant current charging;

(2) TCN and LSTM neural networks were established, and a cross-attention mechanism was employed to fuse the two networks, resulting in the construction of a TCN-LSTM parallel model;

(3) The model was validated using the CALCE dataset, and the TCN-LSTM model achieved an RMSE of 0.0111, showing significant improvement over single-model approaches.

REFERENCES

- [1] A. Oliveira, G. Couto, and P. Pimentel, "Currents of change: Social-environmental valuation of electric ships for sustainable passenger transport," *Econ. Transp.*, vol. 42, p. 100413, Jun. 2025.
- [2] X. Li *et al.*, "A novel capacity allocation method for hybrid energy storage system for electric ship considering life cycle cost," *J. Energy Storage*, vol. 116, p. 116070, Apr. 2025.
- [3] O. Ojo, H. Lang, Y. Kim, X. Hu, B. Mu, and X. Lin, "A Neural Network Based Method for Thermal Fault Detection in Lithium-Ion Batteries," *IEEE Trans. Ind. Electron.*, vol. 68, no. 5, pp. 4068–4078, May 2021.
- [4] W. Zhang, T. Li, W. Wu, N. Ouyang, and G. Huang, "Data-driven state of health estimation in retired battery based on low and medium-frequency electrochemical impedance spectroscopy," *Measurement*, vol. 211, p. 112597, Apr. 2023.

- [5] Y. Ma, C. Shan, J. Gao, and H. Chen, "A novel method for state of health estimation of lithium-ion batteries based on improved LSTM and health indicators extraction," *Energy*, vol. 251, p. 123973, Jul. 2022.
- [6] "State-of-charge estimation of lithium-ion batteries using LSTM and UKF," *Energy*, vol. 201, p. 117664, Jun. 2020.
- [7] L. Ren, J. Dong, X. Wang, Z. Meng, L. Zhao, and M. J. Deen, "A Data-Driven Auto-CNN-LSTM Prediction Model for Lithium-Ion Battery Remaining Useful Life," *IEEE Trans. Ind. Inform.*, vol. 17, no. 5, pp. 3478–3487, May 2021.
- [8] J. Bai, J. Huang, K. Luo, F. Yang, and Y. Xian, "A feature reuse based multi-model fusion method for state of health estimation of lithium-ion batteries," *J. Energy Storage*, vol. 70, p. 107965, Oct. 2023.
- [9] F. Guo *et al.*, "Lithium Battery State-of-Health Estimation Based on Sample Data Generation and Temporal Convolutional Neural Network," *Energies*, vol. 16, no. 24, p. 8010, Dec. 2023.
- [10] H. Zhang, W. Tang, W. Na, P.-Y. Lee, and J. Kim, "Implementation of generative adversarial network-CLS combined with bidirectional long short-term memory for lithium-ion battery state prediction," *J. Energy Storage*, vol. 31, p. 101489, Oct. 2020.